

$$\begin{aligned} & \frac{1}{16\pi^2} \left(-\frac{1}{144} \frac{1}{m_s^2} s_\gamma^2 y_e^{11p} (g_2^2 \overline{y_e^{i2p}} + 12 c_\gamma^2 \overline{y_e^{rp}} \overline{y_e^{i2s}} y_e^{rs}) + \right. \\ & \frac{1}{9} g_2^4 \text{LF}_{3,0}[m_2] \delta_{i1i2} + \frac{1}{6} g_2^4 \text{LF}_{4,-1}[m_2] \delta_{i1i2} - \frac{8}{45} g_2^4 \text{LF}_{5,-2}[m_2] \delta_{i1i2} + \\ & \frac{1}{18} \sum_p g_2^4 \text{LF}_{3,0}[m_\Gamma^p] \delta_{i1i2} - \frac{5}{48} \sum_p g_2^4 \text{LF}_{4,-1}[m_\Gamma^p] \delta_{i1i2} + \frac{2}{45} \sum_p g_2^4 \text{LF}_{5,-2}[m_\Gamma^p] \delta_{i1i2} + \\ & \frac{1}{6} \sum_p g_2^4 \text{LF}_{3,0}[m_q^p] \delta_{i1i2} - \frac{5}{16} \sum_p g_2^4 \text{LF}_{4,-1}[m_q^p] \delta_{i1i2} + \frac{2}{15} \sum_p g_2^4 \text{LF}_{5,-2}[m_q^p] \delta_{i1i2} - \\ & \frac{1}{24} g_2^2 s_\gamma^2 \overline{y_e^{i2p}} y_e^{i1p} \text{LF}_{2,1}[m_\Phi] + \frac{1}{72} (3 g_2^2 s_\gamma^2 \overline{y_e^{i2p}} y_e^{i1p} + 4 g_2^4 \delta_{i1i2}) \text{LF}_{3,0}[m_\Phi] - \\ & \frac{5}{48} g_2^4 \text{LF}_{4,-1}[m_\Phi] \delta_{i1i2} + \frac{2}{45} g_2^4 \text{LF}_{5,-2}[m_\Phi] \delta_{i1i2} + \frac{1}{18} g_2^4 \text{LF}_{3,0}[\tilde{\mu}] \delta_{i1i2} + \\ & \frac{1}{12} g_2^4 \text{LF}_{4,-1}[\tilde{\mu}] \delta_{i1i2} - \frac{4}{45} g_2^4 \text{LF}_{5,-2}[\tilde{\mu}] \delta_{i1i2} + \frac{1}{2} g_1^2 c_\gamma^2 \overline{y_e^{i2p}} y_e^{i1p} \text{LF}_{2,1,0}[m_1, m_e^p] - \\ & g_1^2 c_\gamma^2 \overline{y_e^{i2p}} y_e^{i1p} \text{LF}_{3,1,-1}[m_1, m_e^p] + \frac{1}{2} g_1^2 c_\gamma^2 \overline{y_e^{i2p}} y_e^{i1p} \text{LF}_{4,1,-2}[m_1, m_e^p] + \\ & \frac{1}{96} g_1^2 g_2^2 \text{LF}_{2,1,0}[m_1, m_\Gamma^{i1}] \delta_{i1i2} + \frac{1}{96} g_1^2 g_2^2 \text{LF}_{2,2,-1}[m_1, m_\Gamma^{i1}] \delta_{i1i2} - \\ & \frac{1}{48} g_1^2 g_2^2 \text{LF}_{3,1,-1}[m_1, m_\Gamma^{i1}] \delta_{i1i2} + \frac{1}{96} g_1^2 g_2^2 \text{LF}_{4,1,-2}[m_1, m_\Gamma^{i1}] \delta_{i1i2} + \\ & \frac{1}{96} g_1^2 g_2^2 \text{LF}_{2,1,0}[m_1, m_\Gamma^{i2}] \delta_{i1i2} + \frac{1}{96} g_1^2 g_2^2 \text{LF}_{2,2,-1}[m_1, m_\Gamma^{i2}] \delta_{i1i2} - \\ & \frac{1}{48} g_1^2 g_2^2 \text{LF}_{3,1,-1}[m_1, m_\Gamma^{i2}] \delta_{i1i2} + \frac{1}{96} g_1^2 g_2^2 \text{LF}_{4,1,-2}[m_1, m_\Gamma^{i2}] \delta_{i1i2} - \\ & \frac{1}{96} g_2^4 \text{LF}_{2,1,0}[m_2, m_\Gamma^{i1}] \delta_{i1i2} - \frac{1}{96} g_2^4 \text{LF}_{2,2,-1}[m_2, m_\Gamma^{i1}] \delta_{i1i2} - \frac{5}{48} g_2^4 \text{LF}_{3,1,-1}[m_2, m_\Gamma^{i1}] \delta_{i1i2} + \\ & \frac{1}{32} g_2^4 \text{LF}_{4,1,-2}[m_2, m_\Gamma^{i1}] \delta_{i1i2} - \frac{1}{96} g_2^4 \text{LF}_{2,1,0}[m_2, m_\Gamma^{i2}] \delta_{i1i2} - \frac{1}{96} g_2^4 \text{LF}_{2,2,-1}[m_2, m_\Gamma^{i2}] \delta_{i1i2} - \\ & \frac{5}{48} g_2^4 \text{LF}_{3,1,-1}[m_2, m_\Gamma^{i2}] \delta_{i1i2} + \frac{1}{32} g_2^4 \text{LF}_{4,1,-2}[m_2, m_\Gamma^{i2}] \delta_{i1i2} - \frac{1}{3} g_2^4 \text{LF}_{2,1,0}[m_2, \tilde{\mu}] \delta_{i1i2} - \\ & \frac{1}{3} g_2^4 \text{LF}_{2,2,-1}[m_2, \tilde{\mu}] \delta_{i1i2} - \frac{2}{3} m_2 s_\gamma \tilde{\mu} c_\gamma g_2^4 \text{LF}_{2,2,0}[m_2, \tilde{\mu}] \delta_{i1i2} + \frac{1}{6} g_2^4 \text{LF}_{3,1,-1}[m_2, \tilde{\mu}] \delta_{i1i2} + \\ & \frac{1}{6} g_2^4 \text{LF}_{3,2,-2}[m_2, \tilde{\mu}] \delta_{i1i2} + \frac{1}{3} m_2 s_\gamma \tilde{\mu} c_\gamma g_2^4 \text{LF}_{3,2,-1}[m_2, \tilde{\mu}] \delta_{i1i2} + \\ & \frac{1}{8} g_2^2 (c_\gamma \overline{a_e^{pr}} - s_\gamma \tilde{\mu} \overline{y_e^{pr}}) (c_\gamma a_e^{pr} - s_\gamma \tilde{\mu} y_e^{pr}) \text{LF}_{4,1,-1}[m_\Gamma^p, m_e^r] \delta_{i1i2} - \\ & \frac{1}{6} g_2^2 (c_\gamma \overline{a_e^{pr}} - s_\gamma \tilde{\mu} \overline{y_e^{pr}}) (c_\gamma a_e^{pr} - s_\gamma \tilde{\mu} y_e^{pr}) \text{LF}_{5,1,-2}[m_\Gamma^p, m_e^r] \delta_{i1i2} - \\ & \frac{1}{48} g_1^2 g_2^2 \text{LF}_{2,1,0}[m_\Gamma^{i1}, m_1] \delta_{i1i2} + \frac{1}{96} g_1^2 g_2^2 \text{LF}_{3,1,-1}[m_\Gamma^{i1}, m_1] \delta_{i1i2} + \\ & \frac{1}{48} g_2^4 \text{LF}_{2,1,0}[m_\Gamma^{i1}, m_2] \delta_{i1i2} - \frac{1}{96} g_2^4 \text{LF}_{3,1,-1}[m_\Gamma^{i1}, m_2] \delta_{i1i2} - \\ & \frac{1}{48} g_1^2 g_2^2 \text{LF}_{2,1,0}[m_\Gamma^{i2}, m_1] \delta_{i1i2} + \frac{1}{96} g_1^2 g_2^2 \text{LF}_{3,1,-1}[m_\Gamma^{i2}, m_1] \delta_{i1i2} + \\ & \frac{1}{48} g_2^4 \text{LF}_{2,1,0}[m_\Gamma^{i2}, m_2] \delta_{i1i2} - \frac{1}{96} g_2^4 \text{LF}_{3,1,-1}[m_\Gamma^{i2}, m_2] \delta_{i1i2} + \\ & \frac{3}{8} g_2^2 (c_\gamma \overline{a_d^{pr}} - s_\gamma \tilde{\mu} \overline{y_d^{pr}}) (c_\gamma a_d^{pr} - s_\gamma \tilde{\mu} y_d^{pr}) \text{LF}_{4,1,-1}[m_q^p, m_d^r] \delta_{i1i2} - \\ & \frac{1}{2} g_2^2 (c_\gamma \overline{a_d^{pr}} - s_\gamma \tilde{\mu} \overline{y_d^{pr}}) (c_\gamma a_d^{pr} - s_\gamma \tilde{\mu} y_d^{pr}) \text{LF}_{5,1,-2}[m_q^p, m_d^r] \delta_{i1i2} + \\ & \frac{1}{4} g_2^2 (s_\gamma \overline{a_u^{pr}} - \tilde{\mu} c_\gamma \overline{y_u^{pr}}) (s_\gamma a_u^{pr} - \tilde{\mu} c_\gamma y_u^{pr}) \text{LF}_{2,2,0}[m_q^p, m_u^r] \delta_{i1i2} - \\ & \frac{1}{8} g_2^2 (s_\gamma \overline{a_u^{pr}} - \tilde{\mu} c_\gamma \overline{y_u^{pr}}) (s_\gamma a_u^{pr} - \tilde{\mu} c_\gamma y_u^{pr}) \text{LF}_{3,2,-1}[m_q^p, m_u^r] \delta_{i1i2} - \\ & \frac{1}{4} g_2^2 (s_\gamma \overline{a_u^{pr}} - \tilde{\mu} c_\gamma \overline{y_u^{pr}}) (s_\gamma a_u^{pr} - \tilde{\mu} c_\gamma y_u^{pr}) \text{LF}_{3,1,0}[m_u^r, m_q^p] \delta_{i1i2} - \\ & \frac{1}{4} g_2^2 (s_\gamma \overline{a_u^{pr}} - \tilde{\mu} c_\gamma \overline{y_u^{pr}}) (s_\gamma a_u^{pr} - \tilde{\mu} c_\gamma y_u^{pr}) \text{LF}_{3,2,-1}[m_u^r, m_q^p] \delta_{i1i2} + \\ & \frac{3}{4} g_2^2 (s_\gamma \overline{a_u^{pr}} - \tilde{\mu} c_\gamma \overline{y_u^{pr}}) (s_\gamma a_u^{pr} - \tilde{\mu} c_\gamma y_u^{pr}) \text{LF}_{4,1,-1}[m_u^r, m_q^p] \delta_{i1i2} - \\ & \frac{1}{2} g_2^2 (s_\gamma \overline{a_u^{pr}} - \tilde{\mu} c_\gamma \overline{y_u^{pr}}) (s_\gamma a_u^{pr} - \tilde{\mu} c_\gamma y_u^{pr}) \text{LF}_{5,1,-2}[m_u^r, m_q^p] \delta_{i1i2} - \\ & \frac{5}{24} g_1^2 g_2^2 \text{LF}_{4,1,-2}[\tilde{\mu}, m_1] \delta_{i1i2} - \frac{1}{4} m_1 s_\gamma \tilde{\mu} c_\gamma g_1^2 g_2^2 \text{LF}_{4,1,-1}[\tilde{\mu}, m_1] \delta_{i1i2} + \\ & \frac{1}{6} g_1^2 g_2^2 \text{LF}_{5,1,-3}[\tilde{\mu}, m_1] \delta_{i1i2} + \frac{1}{3} m_1 s_\gamma \tilde{\mu} c_\gamma g_1^2 g_2^2 \text{LF}_{5,1,-2}[\tilde{\mu}, m_1] \delta_{i1i2} + \\ & \frac{2}{3} g_2^4 \text{LF}_{3,1,-1}[\tilde{\mu}, m_2] \delta_{i1i2} + \frac{2}{3} m_2 s_\gamma \tilde{\mu} c_\gamma g_2^4 \text{LF}_{3,1,0}[\tilde{\mu}, m_2] \delta_{i1i2} + \frac{1}{3} g_2^4 \text{LF}_{3,2,-2}[\tilde{\mu}, m_2] \delta_{i1i2} + \\ & \frac{2}{3} m_2 s_\gamma \tilde{\mu} c_\gamma g_2^4 \text{LF}_{3,2,-1}[\tilde{\mu}, m_2] \delta_{i1i2} - \frac{9}{8} g_2^4 \text{LF}_{4,1,-2}[\tilde{\mu}, m_2] \delta_{i1i2} - \\ & \frac{7}{4} m_2 s_\gamma \tilde{\mu} c_\gamma g_2^4 \text{LF}_{4,1,-1}[\tilde{\mu}, m_2] \delta_{i1i2} + \frac{1}{2} g_2^4 \text{LF}_{5,1,-3}[\tilde{\mu}, m_2] \delta_{i1i2} + \\ & m_2 s_\gamma \tilde{\mu} c_\gamma g_2^4 \text{LF}_{5,1,-2}[\tilde{\mu}, m_2] \delta_{i1i2} - \frac{1}{8} g_2^2 \overline{y_e^{i2p}} y_e^{i1p} \text{LF}_{3,1,-1$$